

## SANDOMIERZ, Poland

WMO: 125850

Lat: 50.6967N

Lon: 21.7158E

Elev: 219

StdP: 98.72

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | -15.3      | -12.2 | -17.8                         | 0.8 | -14.6 | -14.9 | 1.1 | -11.7 | 10.2                  | 4.1  | 8.9 | 2.0  | 2.5                      | 30   | 0.527 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.6                    | 29.7            | 20.5 | 27.8 | 19.7 | 26.2 | 18.9 | 21.7                | 27.7 | 20.7 | 26.2 | 19.8 | 24.7 | 2.9                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.6                            | 14.7 | 24.6 | 18.7 | 13.9 | 23.3 | 18.0 | 13.3 | 22.3 | 64.1          | 28.0 | 60.6 | 26.2 | 57.5 | 24.8 | 24.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 8.2           | 7.2  | 6.3 | DB                         | -17.9 | 32.5               | 4.0 | 1.4                                                | -20.8 | 33.5       | -23.2 | 34.3       | -25.5 | 35.1       | -28.4 | 36.1 | (4) |
| (5)               |      |     | WB                         | -18.2 | 23.2               | 4.0 | 0.8                                                | -21.1 | 23.7       | -23.4 | 24.2       | -25.6 | 24.6       | -28.5 | 25.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.9         | -2.4  | -0.8  | 3.3   | 9.5   | 14.3  | 17.5  | 19.6  | 19.0  | 14.1  | 8.9   | 4.0   | -0.7  | (6)  |  |
|                                                                        | DBStd       | 8.80        | 5.23  | 5.15  | 4.59  | 4.37  | 3.84  | 3.47  | 3.20  | 3.06  | 3.54  | 4.25  | 4.42  | 4.84  | (7)  |  |
|                                                                        | HDD10.0     | 1562        | 383   | 301   | 211   | 60    | 7     | 0     | 0     | 0     | 7     | 74    | 185   | 333   | (8)  |  |
|                                                                        | HDD18.3     | 3599        | 642   | 534   | 466   | 267   | 133   | 56    | 24    | 29    | 133   | 293   | 431   | 592   | (9)  |  |
|                                                                        | CDD10.0     | 1162        | 0     | 0     | 3     | 44    | 142   | 227   | 297   | 278   | 129   | 39    | 4     | 0     | (10) |  |
|                                                                        | CDD18.3     | 158         | 0     | 0     | 0     | 1     | 9     | 32    | 63    | 48    | 6     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 1244        | 0     | 0     | 0     | 13    | 87    | 242   | 501   | 361   | 41    | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 290         | 0     | 0     | 0     | 1     | 10    | 49    | 138   | 88    | 3     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.2         | 3.6   | 3.6   | 3.8   | 3.4   | 3.1   | 3.0   | 2.8   | 2.6   | 2.8   | 3.0   | 3.3   | 3.5   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 583         | 28    | 23    | 33    | 44    | 69    | 68    | 95    | 60    | 60    | 44    | 34    | 26    | (15) |  |
|                                                                        | PrecMax     | 791         | 56    | 55    | 71    | 94    | 173   | 157   | 340   | 119   | 117   | 114   | 80    | 56    | (16) |  |
|                                                                        | PrecMin     | 418         | 6     | 9     | 11    | 5     | 32    | 37    | 10    | 6     | 7     | 5     | 0     | 9     | (17) |  |
|                                                                        | PrecStd     | 85          | 12    | 11    | 17    | 22    | 33    | 30    | 67    | 29    | 32    | 29    | 16    | 12    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 9.1   | 12.1  | 17.7  | 24.9  | 28.1  | 30.5  | 32.2  | 31.6  | 27.0  | 21.5  | 15.7  | 10.2  | (19) |  |
|                                                                        |             | MCWB        | 6.8   | 8.2   | 10.6  | 15.5  | 17.9  | 21.2  | 21.1  | 21.3  | 18.4  | 16.1  | 12.0  | 8.0   | (20) |  |
|                                                                        | 2%          | DB          | 7.0   | 9.3   | 14.6  | 21.9  | 25.9  | 28.2  | 30.1  | 29.0  | 24.4  | 19.3  | 13.3  | 8.1   | (21) |  |
|                                                                        |             | MCWB        | 5.3   | 6.6   | 9.0   | 13.5  | 17.1  | 20.2  | 20.5  | 20.5  | 17.6  | 14.9  | 10.6  | 6.6   | (22) |  |
|                                                                        | 5%          | DB          | 5.3   | 7.5   | 12.2  | 19.1  | 23.8  | 26.1  | 28.0  | 27.0  | 22.4  | 17.3  | 11.6  | 6.4   | (23) |  |
|                                                                        |             | MCWB        | 4.1   | 5.5   | 7.8   | 12.1  | 16.5  | 18.9  | 19.9  | 19.6  | 16.5  | 13.9  | 9.6   | 5.2   | (24) |  |
|                                                                        | 10%         | DB          | 3.9   | 5.7   | 9.9   | 16.9  | 21.7  | 24.4  | 26.1  | 25.2  | 20.4  | 15.2  | 10.0  | 5.0   | (25) |  |
|                                                                        |             | MCWB        | 2.9   | 4.2   | 6.7   | 11.0  | 15.4  | 17.9  | 19.1  | 18.5  | 15.6  | 12.7  | 8.6   | 4.0   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 7.2   | 8.7   | 11.3  | 15.9  | 19.7  | 22.2  | 22.8  | 22.8  | 19.8  | 16.9  | 12.4  | 8.8   | (27) |  |
|                                                                        |             | MCDB        | 8.7   | 11.1  | 16.2  | 23.8  | 25.5  | 28.4  | 29.2  | 29.3  | 25.0  | 20.3  | 15.0  | 10.1  | (28) |  |
|                                                                        | 2%          | WB          | 5.5   | 7.1   | 9.6   | 14.3  | 18.1  | 21.0  | 21.7  | 21.4  | 18.2  | 15.5  | 11.1  | 6.7   | (29) |  |
|                                                                        |             | MCDB        | 6.8   | 9.0   | 13.9  | 20.5  | 23.7  | 26.6  | 27.8  | 27.3  | 23.2  | 18.5  | 12.8  | 7.9   | (30) |  |
|                                                                        | 5%          | WB          | 4.2   | 5.6   | 8.3   | 12.8  | 17.0  | 19.8  | 20.8  | 20.3  | 17.1  | 14.2  | 9.8   | 5.4   | (31) |  |
|                                                                        |             | MCDB        | 5.2   | 7.2   | 11.5  | 17.8  | 22.5  | 24.9  | 26.3  | 25.7  | 21.3  | 16.8  | 11.4  | 6.4   | (32) |  |
|                                                                        | 10%         | WB          | 2.9   | 4.2   | 6.9   | 11.6  | 16.0  | 18.6  | 19.9  | 19.3  | 16.1  | 12.8  | 8.6   | 4.1   | (33) |  |
|                                                                        |             | MCDB        | 3.8   | 5.5   | 9.4   | 16.0  | 21.0  | 22.9  | 24.6  | 23.7  | 19.5  | 14.9  | 9.9   | 4.9   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.7   | 5.8   | 7.2   | 9.1   | 9.6   | 9.3   | 9.6   | 9.8   | 8.7   | 7.5   | 5.2   | 4.5   | (35) |  |
|                                                                        |             | MCDBR       | 5.3   | 8.1   | 11.5  | 12.9  | 12.6  | 12.1  | 12.5  | 12.5  | 11.9  | 10.5  | 7.7   | 5.6   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 4.3   | 5.7   | 6.9   | 6.3   | 6.0   | 5.6   | 4.9   | 5.4   | 5.9   | 6.4   | 5.3   | 4.5   | (37) |  |
|                                                                        |             | MCWBR       | 5.1   | 7.6   | 10.6  | 11.9  | 11.5  | 11.0  | 11.0  | 11.6  | 10.8  | 10.0  | 7.1   | 5.4   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.317 | 0.342 | 0.382 | 0.436 | 0.424 | 0.423 | 0.446 | 0.440 | 0.407 | 0.370 | 0.350 | 0.314 | (40) |  |
|                                                                        |             | taud        | 2.252 | 2.237 | 2.211 | 2.115 | 2.211 | 2.245 | 2.192 | 2.209 | 2.261 | 2.336 | 2.319 | 2.309 | (41) |  |
|                                                                        | Edn at Noon | Edn at Noon | 689   | 773   | 808   | 801   | 833   | 836   | 807   | 791   | 774   | 734   | 646   | 638   | (42) |  |
|                                                                        |             | Edh at Noon | 71    | 93    | 115   | 142   | 135   | 132   | 137   | 128   | 109   | 85    | 68    | 59    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 0.81  | 1.56  | 2.73  | 4.09  | 5.09  | 5.46  | 5.14  | 4.63  | 3.27  | 1.94  | 0.90  | 0.61  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.14  | 0.27  | 0.35  | 0.48  | 0.57  | 0.44  | 0.59  | 0.34  | 0.48  | 0.28  | 0.10  | 0.09  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.55 | N/A     | N/A    | +0.93   | N/A   | N/A   | N/A         | N/A     | +102    | +53     |
| (47) Regional (26 neighbors) |  | +0.53 | N/A     | N/A    | +0.85   | +0.44 | +0.41 | N/A         | -167    | +83     | +36     |

Nomenclature: See separate page